

Trading Bot v4 — Web Dashboard Design Brief

A self-contained brief you can hand to a designer (e.g. "Claude Design") cold. Everything they need: what the product is, who uses it, what to show, what to emphasize, what to avoid. No prior conversation context required.

1. Product in one paragraph

This is an **autonomous systematic trading laboratory** run by a single solo-retail operator. It is *not* a chatbot that trades and *not* a robo-advisor. LLMs are used to generate hypotheses, mutate parameters, review research, and write postmortems. A deterministic, hash-locked **risk + ledger + validation + execution kernel** is the only thing that decides what may actually trade. The dashboard is the operator's cockpit: it shows what the deterministic kernel is doing right now, what the research factory is proposing, and what — if anything — requires a human decision.

Current autonomy level is **L2: autonomous research, no autonomous live trading**. The bot may run sandbox experiments and validation packets; it may **not** promote a strategy to real capital. The dashboard must reinforce this — research and trading are visually separated, and any live-trading affordance is gated behind explicit safety flags.

The system trades on **Alpaca paper** today. Equity is real-tracked but the broker is paper. Lanes in play: **ETF Momentum (stocks)**, **Crypto** (capped at 15% of equity), and a **Wheel (options)** sleeve scaffolded but not active.

2. Who uses it

Primary user: the operator (one person, technical, ex-engineer mindset). They check the dashboard:

- **First thing in the morning** — what fired overnight, anything broken, is the daemon healthy, are there any orphaned orders or reconciliation mismatches.
- **Around the open / close** — what trades happened, are positions where they should be, what's the regime, is anything blocked.
- **A few times during the day** — only if something pages them. Otherwise the dashboard should be *boring on purpose*.
- **Weekly** — strategy review: what hypotheses did research propose, what survived BH-FDR multiple testing, what's queued for promotion.

Mental model: they think like an SRE running a service. The cockpit should feel like a production dashboard, not a brokerage app. No gamification, no streaks, no celebratory animations. Numbers, sources, hashes, timestamps.

Secondary user (eventually): the same operator on mobile, glancing at one "is everything OK?" screen.

3. The four product surfaces

The dashboard is one app with four top-level surfaces. The designer's main job is the IA, hierarchy, and tone of each.

Surface	Question it answers	Refresh cadence
A. Right Now	"Is the bot healthy and what is it doing this minute?"	Live (10–25s)
B. Recent Activity	"What happened in the last hours / day?"	1–5 min
C. Strategy Lab	"What is research proposing, and what's the validation state?"	On demand
D. System Health	"Is the plumbing OK — jobs, data freshness, reconciliation, costs, hashes?"	30–60s

A persistent **top bar** is visible on every surface and carries the four pieces of always-on information (see §6).

4. Hard design constraints

These are non-negotiable. They come from the product's safety model, not taste.

1. **The kill switch is sacred.** A single visible affordance, top-right of every page, that halts all new entries. It must be visually obvious, must require a typed reason (no one-click halt), and must show its current state with a color the rest of the UI never uses.
2. **No celebratory styling for P&L.** Green = up, red = down is fine, but no confetti, no 🎉, no gradient hero numbers. Equity is a number, not a brand.
3. **Provenance over polish.** Every non-trivial number should be reachable to a source: a ledger row, a lock file, a timestamp, a strategy version, a hash. Hover or click reveals provenance. The designer should leave room for a small monospace metadata line under most data points.
4. **Research ≠ Trading.** The **Strategy Lab** surface must look visually distinct from **Right Now**. Different background tint or border treatment. It's a research workspace; nothing on it can move money.
5. **Localhost-only aesthetic.** The dashboard binds 127.0.0.1. It's an internal tool; designs that feel like a marketing site (hero sections, illustrations, testimonials) are wrong. Think: Grafana, Datadog, Vercel's internal admin.
6. **Information density is a feature.** The operator wants ~6–10 panels visible without scrolling on a 1440-wide monitor. Don't waste vertical space on oversized cards. Compact > airy.
7. **The pessimistic cost lens is the default.** When showing strategy performance, the default lens is the pessimistic backtest, not the raw or broker-paper lens. The other two lenses are toggleable but secondary.
8. **No invented data.** If a number isn't in the ledger or the snapshot API, it shouldn't be on the screen. The designer should mark mocked numbers in their mockups with a clear `[mock]` label so they don't ship by accident.

5. Visual tone

- **Palette:** neutral dark by default (operator stares at it for hours), with a one-click light mode. Two semantic colors: success/up green, fault/down red. One alert color (amber) for "needs attention but not broken." One reserved color (e.g. magenta) for halt/kill state — nothing else uses it.
 - **Type:** one sans for UI, one monospace for numbers, hashes, ledger seqs, symbols. Tabular numerals everywhere a column has numbers.
 - **Density:** Stripe Dashboard / Linear / Grafana density, not Notion density.
 - **Motion:** minimal. Fade-in on new ledger rows, no slide-ins, no parallax.
 - **Iconography:** lucide / heroicons line set, not filled. No emojis.
 - **No screenshots of charts as decoration.** Every chart shown must be live.
-

6. Persistent top bar (every page)

Single row, always visible. Left to right:

1. **System status pill** — one of `running` / `degraded` / `halted` / `down` . Color-coded; the magenta-reserved color is the `halted` state.
 2. **Equity + day P&L** — small, monospace. `$54,317.42 ▲ +$184 (+0.34%)` . Click → equity curve drawer.
 3. **Active lanes** — chips: `ETF Momentum` `Crypto` `Wheel (off)` . Each chip shows its current exposure as a thin bar inside the chip.
 4. **Risk profile** — `safe` / `neutral` / `aggressive` . Click → profile drawer.
 5. **Big red KILL** button — top right. Always visible. Click → modal with typed reason; while halted, it becomes a green **Resume** button and a magenta banner spans the page width.
-

7. Surface A — Right Now

The default landing. Three-column layout on wide screens, one column on mobile.

Column 1 — Posture

- **Strategy mode** card: which strategies are armed, paused, or research-only.
- **Regime** card: a single tag (`bullish` / `chop` / `risk-off`) sourced from the regime detector, with the macro signals that drove it.
- **Risk caps** card: 4–6 rows, each a horizontal bar showing `used` / `cap` for: account exposure, crypto exposure, lane caps, PDT count, strategy-level cap. Bars turn amber at 80% and red above cap.

Column 2 — Positions and orders

- **Positions table:** symbol, lane, qty, entry, mark, P&L \$, P&L %, classification (`bot` / `external` / `manual` / `unknown`), stop, age. The `unknown` classification must be visually loud — the kernel halts

on unknown in Phase 2+, so the operator needs to see it instantly. Sortable. Expandable rows reveal: strategy version that opened it, ledger order_uid, last fill timestamp, drift vs. expected.

- **Open orders** table: symbol, side, qty, type, status, idempotency key, age, client_order_id. Strikethrough on canceled. Amber on stuck > 60s.

Column 3 — Action required + activity feed

- **Action required** stack (often empty — that's the goal). Each card has a title, a one-line cause, and one or two CTAs. Examples of what shows up:
 - "Unknown position detected: AAPL × 50. Classify or close."
 - "Reconciliation mismatch on order ord_01HK... — 1 fill local, 2 broker."
 - "Halt active: 'manual_operator_halt' — set by you at 09:14."
 - "Validation lock loosened — 7-day cooldown until 2026-05-22."
- **Activity feed**: live tail of ledger events, newest at top, ~50 visible. Each line: timestamp · event type · one-line payload · ledger seq. Filterable by lane and event type.

Footer of this surface

- **Equity curve**: line chart, range selector `1w / 1m / 3m / ytd / all`. Overlaid markers for kill events, profile changes, and policy-lock changes.
- **Exposure donut**: by lane, summing to 100% of equity. Cash is a slice.

8. Surface B — Recent Activity

For the morning / end-of-day pass. Single column, reverse-chronological.

- **Daily digest** at top: a generated 1-screen summary of the last 24h — account delta, jobs run, fills, orders, halts, mutations proposed. Mirrors `bot digest`.
- **Decision activity**: each entry is one decision the system made (entry, exit, skip, mutation accepted, mutation rejected). Each row: time, strategy version, symbol, action, reason summary, ledger link. Expanding reveals the structured decision record (inputs, gates evaluated, outcome).
- **Lessons**: short notes the system / operator wrote about specific decisions. This is a markdown-ish stream, not a table. Tagged by lane.
- **Last scan**: the most recent scan cycle, showing how many symbols were considered, how many passed each gate, which were submitted.

9. Surface C — Strategy Lab

Visually distinct. Different background tint. Header reads "RESEARCH — nothing on this page can move capital."

- **Strategy registry table**: every registered strategy version, with state (`research_only` / `paper` / `live` / `retired`), tier of last validation artifact (1/2/3), pessimistic Sharpe, deflated Sharpe, probability of overfit, last run, hash.

- **Per-strategy drilldown:**
 - Validation tiers achieved with timestamps and links to the artifacts.
 - Walk-forward folds chart (5+ folds + 30% locked holdout).
 - Parameter plateau heatmap.
 - Mutation log: what was proposed, what BH-FDR-survived, what was rejected.
 - Failure memory: rejected variants (90-day rolling).
 - **Hypothesis intake:** a form to submit a new strategy hypothesis. Three submission modes: `draft` (no run), `intake` (adversarial review), `mutate` (enumerate from hash-locked search space). Submission triggers an async research run; results land back here.
 - **LLM spend:** line chart of cost by role (judges on Opus, reviewers on Sonnet). Daily and monthly. Used to keep research from running away.
 - **Promotion queue:** strategies that have passed Tier-3 validation and are awaiting human sign-off for promotion. This is the **only** affordance in the lab that touches the trading kernel, and it must require typed approval.
-

10. Surface D — System Health

For SRE-mode glancing. Two-column.

Column 1 — The plumbing

- **Job scheduler:** every APScheduler job — name, schedule, last run, last duration, next run, last status. Failed runs in red.
- **Data freshness:** per-lane watermarks (`stocks_eod` , `crypto_1m` , etc.). Each row: source, last tick, expected cadence, lag, OK/stale flag.
- **Reconciliation:** last reconciliation run, mismatches found, count of unresolved. Mismatches link to detail rows in the ledger.
- **Drift monitor:** 20-trade rolling drift between expected and realized fill prices. Sparkline + current value vs. threshold.

Column 2 — Trust and accounting

- **Cost model:** current per-trade cost estimate in all three lenses (raw / broker_paper / pessimistic).
 - **Policy locks:** list of the 9 `.lock` files with version, last change, signer, and the master `HASHES` manifest hash. Status pill per lock: `verified` / `mismatch` . Mismatch is a page-level emergency.
 - **Persona files:** 8 personas, each with version + hash status. Same treatment as locks.
 - **Halt history:** every halt and resume in the last 30 days. Reason, operator, ledger seq, duration.
 - **Ledger health:** row counts per table, last seq, last hash, chain verify status. A green check for "chain verified at HH:MM:SS" is the single most reassuring thing on the dashboard.
 - **Daemon heartbeat:** last beat, uptime, host, PID.
-

11. Information not to put anywhere

The designer should *resist* adding these even if they seem useful — they violate the product's safety model.

- **One-click strategy promote-to-live.** Promotion is a multi-step, human-signed process. Don't surface it as a button.
- **Suggested trades or "AI picks".** Hypotheses live in the lab; surfacing them on Right Now would be misleading.
- **A composite "AI confidence" score.** Calibration is per-strategy; a single hero number is a lie.
- **Charts of LLM responses or thinking.** No conversation transcripts in the UI. Decisions show their *inputs* and *outcome*, not the prose.
- **Marketing affordances:** changelog modals, "what's new", onboarding carousels, feature announcements, achievements.

12. Concrete data shapes the designer can mock with

The backend exposes JSON via `/api/status`, `/api/snapshot`, `/api/equity-curve`, and per-fragment HTMX endpoints. Approximate shapes:

```
// /api/status – drives the top bar + Right Now posture
{
  "system_state": "running" | "degraded" | "halted" | "down",
  "halted": { "active": false, "reason": null, "since": null, "operator": null },
  "risk_profile": "safe" | "neutral" | "aggressive",
  "account": {
    "equity": 54317.42,
    "cash": 12104.81,
    "day_pl_abs": 184.0,
    "day_pl_pct": 0.0034,
    "buying_power": 24209.62
  },
  "lanes": [
    { "name": "ETF Momentum", "enabled": true, "exposure_pct": 0.41 },
    { "name": "Crypto", "enabled": true, "exposure_pct": 0.1502, "cap_pct": 0.15
  },
    { "name": "Wheel", "enabled": false, "exposure_pct": 0.0 }
  ],
  "kill_switches": [
    { "name": "manual_operator_halt", "active": false },
    { "name": "crypto_cap_breach", "active": false },
    { "name": "data_staleness", "active": false }
    // ... 8 total
  ],
  "boot_check": { "ok": true, "hash_verified_at": "2026-05-15T13:02:11Z" }
}
```

```
// /api/snapshot – drives Right Now + Recent Activity panels
{
  "positions": [
    {
      "symbol": "SPY", "lane": "ETF Momentum", "qty": 28, "entry": 521.10,
      "mark": 524.88, "pl_abs": 105.84, "pl_pct": 0.00725,
      "classification": "bot",
      "stop": 514.50,
      "opened_at": "2026-05-12T13:42:00Z",
      "order_uid": "ord_01HK6T4S...",
      "strategy_version": "ETF_MOMENTUM_v1#a3f2"
    }
  ],
  "open_orders": [ /* same shape as positions, with side/status/client_order_id */ ],
  "regime": { "label": "chop", "since": "2026-05-13", "signals": [/*...*/] },
  "risk_caps": [
    { "name": "account_exposure", "used": 0.62, "cap": 0.85 },
    { "name": "crypto", "used": 0.15, "cap": 0.15 },
    { "name": "pdt_count", "used": 1, "cap": 3 }
  ],
  "activity": [
    { "ts": "13:02:11Z", "seq": 28411, "type": "fill", "msg": "SPY filled 28 @ 521.10" },
    { "ts": "12:58:02Z", "seq": 28410, "type": "submit", "msg": "submit SPY 28 mkt" }
  ],
  "action_required": [
    { "id": "ar_unknown_pos_1", "severity": "high",
      "title": "Unknown position: AAPL × 50",
      "cta": [{ "label": "Classify", "href": "/positions/AAPL" },
              { "label": "Close", "href": "/positions/AAPL/close" } ] }
  ]
}
```

```
// /api/equity-curve?range=1m
{
  "range": "1m",
  "points": [{ "ts": "2026-04-15T20:00:00Z", "equity": 50124.10 }, /*...*/],
  "markers": [
    { "ts": "2026-04-22T14:01:00Z", "kind": "halt", "label": "manual halt" },
    { "ts": "2026-05-01T13:30:00Z", "kind": "profile", "label": "safe → neutral" },
    { "ts": "2026-05-13T19:00:00Z", "kind": "lock", "label": "validation lock v3" }
  ]
}
```

The full set of panel names the current backend already populates (so the designer knows what's available without inventing): `action_required`, `header`, `strategy_mode`, `regime`, `kpi`, `risk`, `exposure`, `equity`, `orders`, `activity_feed`, `decision_activity`, `lessons`, `last_scan`, `opportunities`, `stats`, `lab_evolution`, `calibrator`, `threshold_overrides`, `intel_pool`, `proposals`, `llm_spend`, `wheel`, `role_health`, `scheduled`, `freshness`, `email_firehose`, `process_registry`, `sidebar_status`, `halts`, `portfolio_rail`, `macro_alloc`, `node_drilldown`.

13. Responsive behavior

- **≥1440 wide:** 3-column Right Now, 2-column System Health.
- **1024–1439:** 2 columns; Action Required moves above Positions.
- **<1024 (tablet / phone):** single column, surface tabs become a bottom nav, the persistent top bar collapses to: status pill, equity, kill button.

A glanceable mobile view of *just the top bar* + *Action Required* is the critical mobile experience. The operator does not need a full mobile UI.

14. Accessibility

- WCAG AA on all text + chart annotations. Don't rely on color alone; pair every red/green with an arrow or symbol.
 - Keyboard: **g r** go to Right Now, **g a** activity, **g l** lab, **g s** system. **?** opens shortcut help. Halt requires explicit click + typed reason — no keyboard shortcut to kill.
 - Reduced motion respected for all transitions.
-

15. What to deliver

For the designer to return:

1. **High-fidelity mocks** of: - The persistent top bar in three states: **running**, **degraded**, **halted**. - Surface A (Right Now) — wide and tablet widths. - Surface C (Strategy Lab) — landing + one strategy drilldown. - The halt modal (the most safety-critical interaction).
2. **A component inventory:** status pill, risk-cap bar, classification tag, ledger-seq chip, lens toggle, tier badge, hash-verified check.
3. **Two themes:** dark default + light. Same components.
4. **A short rationale** (≤1 page) explaining choices that diverge from this brief, so the operator can decide whether to accept the divergence.

Do **not** deliver: marketing pages, onboarding, signup, billing, notification settings UI, mobile apps. Out of scope.

16. One-line summary for the designer

A solo-operator cockpit for a deterministic trading kernel with an LLM research factory attached. SRE-grade density, provenance on every number, a sacred kill switch, and a visually quarantined research lab. Boring on purpose when the bot is healthy.